JEE Main Online Exam 2020

Questions & Solutions

07th January 2020 | Shift - II

PHYSICS

Q.1 A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will double after a time :

(1) $\frac{3mv_0}{qE_0}$ (2) $\frac{\sqrt{2}mv_0}{qE_0}$ (3) $\frac{\sqrt{3}mv_0}{qE_0}$ (4) $\frac{2mv_0}{qE_0}$

Ans. **[3]**

$\Rightarrow V_x^2 + V_y^2 = V^2$

$\Rightarrow V_0^2 + V^2 = 4V_0^2$

$\Rightarrow V = \sqrt{3}V_0$

Velocity in the direction of $\vec{E}$ will change

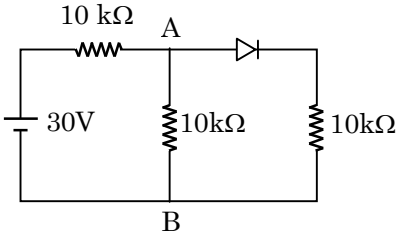
$u_x = 0$

$v_x = \sqrt{3}V_0$

$a_x = \frac{qE_0}{m}$

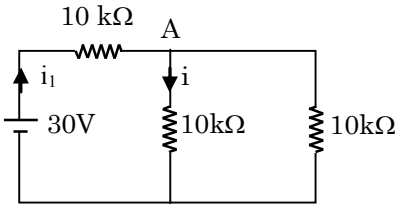
$t = \frac{\sqrt{3}mV_0}{qE_0}$

Q.2 In the figure, potential difference between A and B is :



(1) 10V (2) zero (3) 5 V (4) 15 V

Ans. **[1]**



$$i_1 = \frac{30}{15\times10^3}$$

$i_1 = 2 \times 10^{-3}$

$$i_1 = \frac{i_1}{2} = 10^{-3}$$

$\Delta V = iR = 10^{-3} \times 10 \times 10^3 = 10V$

Q.3 An emf of 20 V is applied at time t = 0 to a circuit containing in series 10 mH inductor and 5Ω resistor. The ratio of the currents at time t = ∞ and at t = 40 s is close to: (Take $e^2 = 7.389$)

(1) 1.15 (2) 1.06 (3) 1.46 (4) 0.84

Ans. **[2]**

$i = i_0(1 - e^{-Rt/L})$

$$i = i_0\left(1 - e^{\frac{-5\times40}{10\times10^{-3}}}\right)$$

$$\frac{i_0}{i} \approx 1$$

Then most appropriate option is 1.06

Q.4 In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is -

(1) 6.9 mm (2) 4.9 mm (3) 3.9 mm (4) 5.9 mm

Ans. **[4]**

$$\beta = \frac{\lambda D}{d}$$

$$= \frac{1.5\times589\times10^{-9}}{0.15\times10^{-3}}$$

= 5.9 mm

Q.5 A box weighs 196 N on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take g = 10 ms^{-2} at the north pole and the radius of the earth = 6400 km)

(1) 194.32 N (2) 195.66 N (3) 195.32 N (4) 194.66 N

Ans. **[3]**

$W = Mg - M\omega^2R$

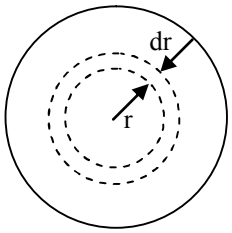
$$= 196 - (19.6)\left(\frac{2\pi}{24\times3600}\right)^2 \times 6400\times10^3$$

$$\boxed{W = 195.32}$$

Q.6 Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is-

(1) $2\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$ (2) $\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$ (3) $2\pi a^4\left(\frac{aA}{4}+\frac{B}{5}\right)$ (4) $2\pi a^4\left(\frac{A}{4}+\frac{B}{5}\right)$

Ans. **[1]**



$dI = dmr^2$

$= \sigma 2\pi r(dr)r^2$

$$\int dI = \int_0^a 2\pi r^3(A + Br)\,dr$$

$$I = 2\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$$

Q.7 In a building there are 15 bulbs of 45W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be -

(1) 15 A (2) 10 A (3) 25 A (4) 20 A

Ans. **[4]**

Total Power $= 15 \times 45 + 15 \times 100 + 15 \times 10 + 2 \times 10^3 = VI$

$$I = \frac{4325}{V} = \frac{4325}{220}$$

$= 19.66$

$I \approx 20A$

Q.8 A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears 2 beats/sec. The oscillation frequency of each tuning fork is ν_0 = 1400 Hz and the velocity of sound in air is 350 m/s. The speed of each tuning fork is close to -

(1) $\frac{1}{4}$ m/s (2) $\frac{1}{2}$ m/s (3) $\frac{1}{8}$ m/s (4) 1 m/s

Ans. **[1]**



observer

$$f_1 = \left(\frac{c}{c-v}\right)f_0$$

$$f_2 = \left(\frac{c}{c+v}\right)f_0$$

beat frequency = $f_1 - f_2$

$$= cf_0\left(\frac{1}{c-v} - \frac{1}{c+v}\right)$$

$$= cf_0\left(\frac{2v}{c^2 - v^2}\right)$$

$c >> v$

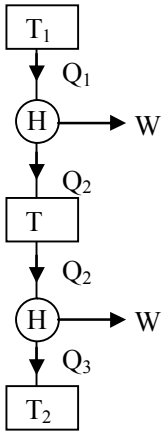
$$= \frac{2f_0 v}{c} = 2$$

$$\boxed{v = \frac{1}{4}\,m/s}$$

Q.9 Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperature, T_1 and T_2. The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2. T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2, if both the engines perform equal amount of work ?

(1) $T = \sqrt{T_1 T_2}$ (2) $T = \frac{T_1 + T_2}{2}$ (3) $T = \frac{2T_1 T_2}{T_1 + T_2}$ (4) $T = 0$

Ans. **[2]**



$$\frac{Q_2}{Q_1} = \frac{T}{T_1} \quad \text{..... (i)}$$

$$\frac{Q_3}{Q_1} = \frac{T_2}{T} \quad \text{..... (ii)}$$

$$\frac{Q_3}{Q_1} = \frac{T_2}{T_1} \quad \text{..... (iii)}$$

work is same

$Q_1 - Q_2 = Q_2 - Q_3$

$\Rightarrow Q_1 + Q_3 = 2Q_2$

$\Rightarrow \frac{T_1}{T} + \frac{T_2}{T} = 2$

$$\boxed{T = \frac{T_1 + T_2}{2}}$$

Q.10 The activity of a radioactive sample falls from 700 s^{-1} to 500s^{-1} in 30 minutes . Its half life is close to -

(1) 66 min (2) 52 min (3) 62 min (4) 72 min

Ans. **[3]**

$A = A_0 e^{-\lambda t}$

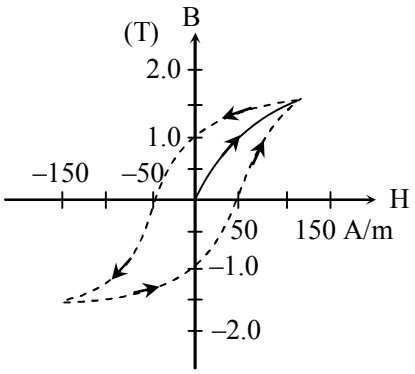
$500 = 700\, e^{-\lambda \times 30}$

$$\ln\frac{7}{5} = \lambda \times 30$$

$$T = \frac{\ln 2}{\lambda} = \frac{30}{\ln\frac{7}{5}} \times \ln 2$$

$$\boxed{T_{Half} \approx 62\,min}$$

Q.11

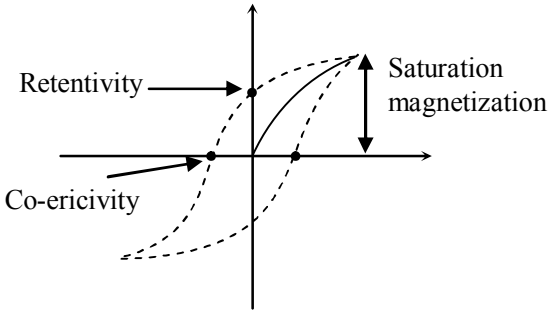


The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are -

(1) 1.0 T, 50A / m and 1.5 T (2) 1.5 T, 50 A/m and 1.0 T

(3) 1.5 T, 50 A/m and 1.0 T (4) 150 A/m, 1.0 T and 1.5 T

Ans. **[1]**



Retentivity = 1.0 T

Co-ercivity = 50 A/m

Saturation = 1.5 T

Q.12 The electric field of a plane electromagnetic wave is given by –

$\vec{E} = E_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz + \omega t)$

At t = 0, a positively charged particle is at the point (x,y, z) = $\left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at (t = 0) is $v_0\hat{k}$, the force acting on it due to the wave is -

(1) parallel to $\hat{k}$ (2) antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (3) zero (4) parallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Ans. **[2]**

$\vec{E} = +E_0\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)\cos\pi \quad (\text{at } t = 0)$

Force due to magnetic field is in direction $q(\vec{v}\times\vec{B})$ & $\vec{v} \parallel \hat{k}$

hence

$\vec{F} = q(\vec{E} + \vec{V}\times\vec{B})$

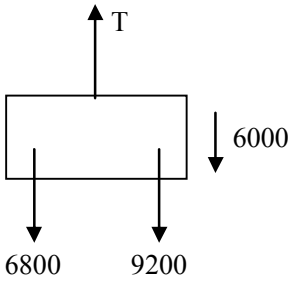
F is in direction of $\vec{E}$

i.e. $-\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

Q.13 An elevator in a building can carry a maximum of 10 person, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator (g = 10 m/s^2) must be at least :

(1) 56300 W (2) 48000 W (3) 66000 W (4) 62360 W

Ans. **[3]**



T = 6800 + 9200 + 6000

= 22000 N

Power = TV = 22000 × 3

= 66000 W

Q.14 A planar loop of wire rotates in a uniform magnetic field. Initially, at t = 0, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at -

(1) 2.5 s and 7.5 s (2) 2.5 s and 5.0 s (3) 5.0 s and 7.5 s (4) 5.0 s and 10.0 s

Ans. **[2]**

$\phi = \vec{B}\cdot\vec{A} = BA\cos\omega t$

$|\varepsilon| = BA\omega\sin\omega t$

ε_{max} when $\omega t = \frac{\pi}{2} \Rightarrow t = 2.5$ sec

ε_{min} when $\omega t = \pi \Rightarrow t = 5$ sec

Q.15 An electron (of mass m) and a photon have the same energy E in the range of a few eV. The ratio of the de-Broglie wavelength associated with the electron and the wavelength of the photon is (c = speed of light in vaccum)

(1) $\frac{1}{c}\left(\frac{2E}{m}\right)^{\frac{1}{2}}$ (2) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (3) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (4) $c(2mE)^{\frac{1}{2}}$

Ans. **[2]**

$\lambda_{De-broglie} = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$

$\lambda_{Photon} = \frac{hc}{E}$

Ratio = $\frac{1}{c}\sqrt{\frac{E}{2m}}$

Q.16 An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocity of fluid in this pipe is :

(1) $\frac{9}{16}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{3}{4}$ (4) $\frac{81}{256}$

Ans. **[1]**

$A_1v_1 = A_2v_2$

$\frac{v_{min}}{v_{max}} = \frac{A_{min}}{A_{max}}$

$\frac{v_{min}}{v_{max}} = \left(\frac{4.8}{6.4}\right)^2$

$= \frac{9}{16}$

Q.17 The dimension of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vaccum, is :

(1) ML^2T^{-2} (2) MLT^{-2} (3) $ML^{-1}T^{-2}$ (4) ML^2T^{-1}

Ans. **[3]**

$\frac{B^2}{2\mu_0}$ = Energy per unit volume

Dimension $\Rightarrow \frac{ML^2T^{-2}}{L^3}$

$\Rightarrow ML^{-1}T^{-2}$

Q.18 A thin lens made of glass (refractive index = 1.5) of focal length f = 16 cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_l, then the ratio f_l / f is closed to the integer

(1) 1 (2) 17 (3) 9 (4) 5

Ans. **[3]**

$$\frac{1}{f} = \left(\frac{\mu_2 - \mu_1}{\mu_1}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{1.5-1}{1}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{.... (i)}$$

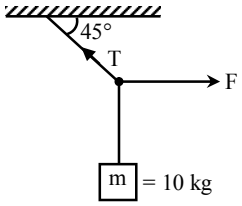
$$\frac{1}{f_1} = \left(\frac{1.5-1.42}{1.42}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{.... (ii)}$$

$$\frac{f_1}{f} = \frac{0.5}{0.056} \approx 9$$

Q.19 A mass of 10 kg is suspended by a rope of length 4m, from the ceiling, A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals : (Take g = 10 ms^{-2} and the rope to be massless) -

(1) 90 N (2) 100 N (3) 70 N (4) 75 N

Ans. **[2]**



T cos 45° = 100

T sin 45° = 100 = F

$\boxed{F = 100N}$

Q.20 Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2. If $\frac{C_P}{C_V} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by -

(1) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$ (2) $\frac{1}{2}$ (3) $\left(\frac{1}{2}\right)^{\gamma}$ (4) 2

Ans. **[BONUS] (NTA answer is [1])**

$\tau \propto \frac{V}{\sqrt{T}}$ ..... (i)

$TV^{\gamma-1} = \text{constant}$ ..... (ii)

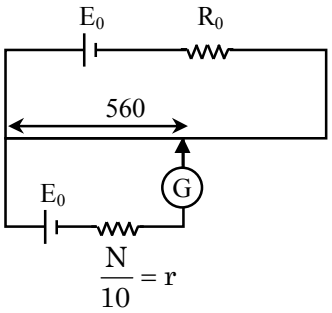
$\tau \propto \frac{V}{V^{\frac{-\gamma+1}{2}}}$

$\tau \propto V^{\frac{1+\gamma}{2}}$

$\frac{\tau_2}{\tau_1} = (2)^{\frac{\gamma+1}{2}}$

Q.21 The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10}\Omega$, where N is an integer then value of N is ..... .

Ans. **[12]**



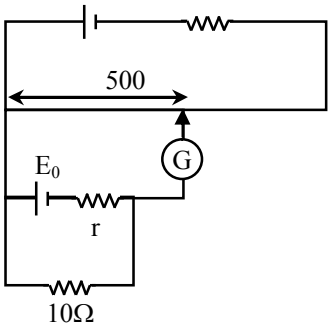
E = 560 x ..... (i)

x → potential gradient

E – ir = 500x

$\frac{10E}{10+r} = 500x$ ..... (ii)

(i) ÷ (ii)



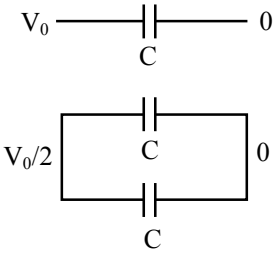
$r = 1.2\ \Omega$

$$\frac{N}{10} = 1.2$$

$$\boxed{N = 12}$$

Q.22 A 60 pF capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is connected to another uncharged 60pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ) ..... .

Ans. **[6]**



$$\Delta E = \frac{1}{2}CV_0^2 - 2\left(\frac{1}{2}C\left(\frac{V_0}{2}\right)^2\right)$$

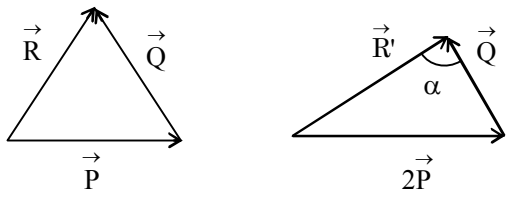
$$= \frac{CV_0^2}{4}$$

$$= 6 \times 10^{-9} = 6\ \text{n J}$$

Q.23 The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is, ...

(1) (2) (3) (4)

Ans. **[90]**



$\vec{P}+\vec{Q}=\vec{R}$

$P^2 + Q^2 + 2PQ\cos\theta = R^2 = P^2$

$2P\cos\theta + Q = 0$ …(1)

$\tan\alpha = \dfrac{2P\sin\theta}{2P\cos\theta + Q}$ …(2)

From (1) & (2)

$\alpha = 90^\circ$

Q.24 M grams of steam at 100° C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is ..... .

Ans. **[40]**

M gm steam 100ºC ↓ M×540 → M gm water 100ºC ↓ MSΔT =M × 1 × 60 = 60 M → M gm water 40ºC

Total heat = 600 M

+

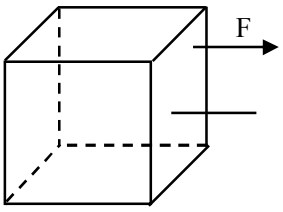
M gm ice 0ºC ↓ 200×80 =16000 → 200 gm water 0ºC ↓ 200×1×40 =8000 → 200 gm water 40ºC

Total heat = 24000 M

600 M = 240000

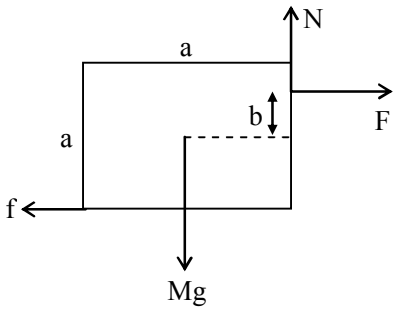
M = 40 gm

Q.25



Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value of $100\times\dfrac{b}{a}$ for box not to topple before moving is ..... .

Ans. **[50]**



Torque about C.O.M.

$$Fb + f\frac{a}{2} = N\frac{a}{2} \qquad \ldots(1)$$

$$F \le f = \mu N \qquad \ldots(2)$$

$$N = Mg \qquad \ldots(3)$$

$$\mu N\left(b + \frac{a}{2}\right) = Mg\ \frac{a}{2}$$

$$\frac{b}{a} = 0.75$$

which is not possible

Hence, answer should be $\frac{b}{a} = 0.5$ (maximum possible)

Ans. 50 $\left(\because 100\frac{b}{a} = 50\right)$

JEE Main Online Exam 2020

Questions & Solutions

7th January 2020 | Shift - II

CHEMISTRY

Q.1 Among the statements(a)-(d), the incorrect ones are :

(a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments

(b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{eg}^4\ e_{eg}^2$

(c) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3-}$

(d) If the Δ_0 for an octahedral complex of Co(III) is 18,000 cm^{-1}, the Δ_t for its tetrahedral complex with the same ligand will be 16,000 cm^{-1}

(1) (c) and (d) only (2) (a) and (d) only (3) (a) and (b) only (4) (b) and (c) only

Ans. **[2]**

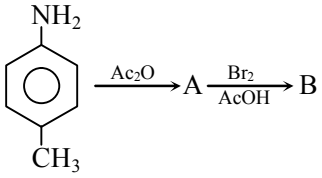
Sol. Strong field ligands have high pairing energy.

For strong field ligand : $-\Delta_0 = \Delta_t \times 1.125$

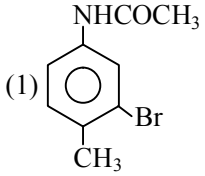
$$\frac{18000}{1.125} = \Delta_t$$

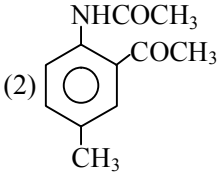
$$16000 = \Delta_t$$

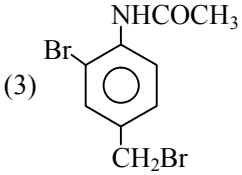
Q.2 If the following reaction sequence,

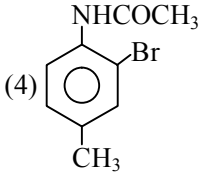


the major product B is :

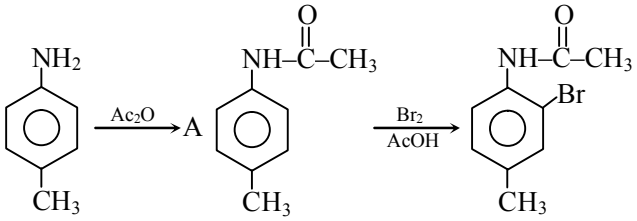
(1) 

(2) 

(3) 

(4) 

Ans. **[4]**

Sol. 

Q.3 Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non volatile solute are together sealed in a container. Over time :

(1) the volume of the solution decreases and the volume of the solvent increases

(2) the volume of the solution and the solvent does not change

(3) the volume of the solution increases and the volume of the solvent decreases

(4) the volume of the solution does not change and the volume of the solvent decreases

Ans. **[3]**

Sol. Lowering of V.P in second Beaker will take place.

Q.4 Among statement (a) – (d), the correct ones are :

(a) Decomposition of hydrogen peroxide gives dioxygen.

(b) Like hydrogen perioxide, compound, such as $KClO_3$, $Pb(NO_3)_2$ and $NaNO_3$ when heated liberate dioxygen.

(c) 2-Ethylanthraquinone is useful for the industrial preparation of hydrogen peroxide.

(d) Hydrogen peroxide is used for the manufacture of sodium perborate.

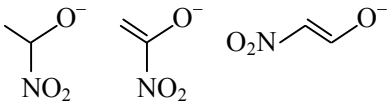
(1) (a), (c) and (d) only (2) (a), (b), (c) and (d)

(3) (a) and (c) only (4) (a), (b) and (c) only

Ans. **[2]**

Sol. Theory Based

Q.5 The correct order of stability for the following alkoxides is :

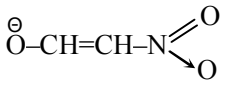


(A) (B) (C)

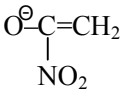
(1) (C) > (A) > (B) (2) (B) > (C) > (A) (3) (B) > (A) > (C) (4) (C) > (B) > (A)

Ans. **[4]**

Sol.



(stable by –M of NO_2)



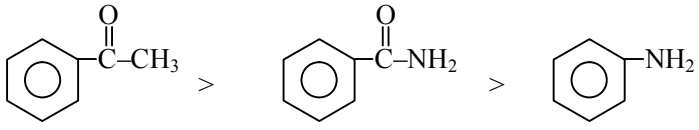
(stable by Resonance)

Q.6 A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvent, hexane : ethyl acetate (20 : 80), the sequence of obtained compounds is :

(1) (B), (C) and (A) (2) (C), (A) and (B) (3) (A), (B) and (C) (4) (B), (A) and (C)

Ans. **[2]**

Sol.



Acetophenone (C) > Acetannilide (A) > Anniline (B)

Anniline has higher viscosity due to efficient H-Bonding.

Q.7 In the following reactions, products (A) and (B), respectively, are :

$NaOH + Cl_2 \rightarrow (A)$ + side products
(hot and conc.)

$Ca(OH)_2 + Cl_2 \rightarrow (B)$ + side products
(dry)

(1) $NaClO_3$ and $Ca(ClO_3)_2$ (2) $NaClO_3$ and $Ca(OCl)_2$
(3) NaOCl and $Ca(OCl)_2$ (4) NaOCl and $Ca(ClO_3)_2$

Ans. **[2]**

Sol. $6NaOH + 3Cl_2 \longrightarrow 5NaCl + NaClO_3 + 3H_2O$

$2Ca(OH)_2 + Cl_2 \longrightarrow Ca(OCl)_2 + CaCl_2 + H_2O$

Q.8 The ammonia (NH_3) released on quantitative reaction of 0.6 g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by :

(1) 200 ml of 0.4 N HCl (2) 200 ml of 0.2 N HCl
(3) 100 ml of 0.1 N HCl (4) 100 ml of 0.2 N HCl

Ans. **[4]**

Sol. 2 × moles of urea = moles of NH_3

moles of NH_3 = moles of HCl

$\frac{100 \times 0.2}{1000}$ = moles of HCl = 0.02 mole of HCl

Q.9 For the following reactions

$CH_3CH_2CH_2Br + Z^{\ominus} \xrightarrow{k_s,\ \text{subtitution}} CH_3CH_2CH_2Z + Br^{\ominus}$

$CH_3CH_2CH_2Br + Z^{\ominus} \xrightarrow{k_e,\ \text{elimination}} CH_3CH{=}CH_2 + HZ + Br^{\ominus}$

where,

$Z^{\ominus} = CH_3CH_2O^{\ominus}$ (A) or $H_3C{-}\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}{-}O^{\ominus}$ (B),

k_s and k_e, are, respectively, the rate constants for substitution and elimination, and $\mu = \frac{k_s}{k_e}$, the correct option is _____.

(1) $\mu_B > \mu_A$ and $k_e(A) > k_e(B)$ (2) $\mu_B > \mu_A$ and $k_e(B) > k_e(A)$
(3) $\mu_A > \mu_B$ and $k_e(B) > k_e(A)$ (4) $\mu_A > \mu_B$ and $k_e(A) > k_e(B)$

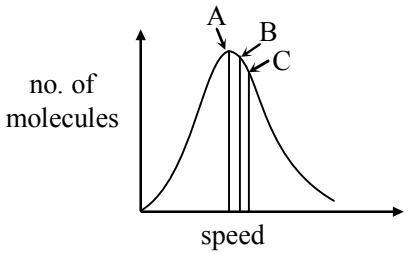
Ans. **[3]**

Sol. Bulkier the base, elimination will be favoured

$\mu_A = \frac{k_{s_A}}{k_{e_A}}$ (For A : $k_s > k_e$)

$\mu_B = \frac{k_{s_B}}{k_{e_B}}$ (For B : $k_s < k_e$)

Q.10 Identify the correct labels of A, B and C in the following graph from the option given below :



Root mean square speed (V_{rms}); most probable speed (V_{mp}); Average speed (V_{av})

(1) A – V_{rms}; B – V_{mp}; C-V_{av} (2) A-V_{mp}; B-V_{av}; C-V_{rms}

(3) A-V_{mp}; B-V_{rms}; C-V_{av} (4) A-V_{av}; B-V_{rms}; C-V_{mp}

Ans. **[2]**

Sol. $V_{rms} > V_{avg} > V_{mps}$

Q.11 The equation that is incorrect is :

(1) $(\Lambda^0_m)_{NaBr} - (\Lambda^0_m)_{NaCl} = (\Lambda^0_m)_{KBr} - (\Lambda^0_m)_{KCl}$

(2) $(\Lambda^0_m)_{H_2O} = (\Lambda^0_m)_{HCl} + (\Lambda^0_m)_{NaOH} - (\Lambda^0_m)_{NaCl}$

(3) $(\Lambda^0_m)_{NaBr} - (\Lambda^0_m)_{NaI} = (\Lambda^0_m)_{KBr} - (\Lambda^0_m)_{NaBr}$

(4) $(\Lambda^0_m)_{KCl} - (\Lambda^0_m)_{NaCl} = (\Lambda^0_m)_{KBr} - (\Lambda^0_m)_{NaBr}$

Ans. **[3]**

Sol. $\Lambda^0_m NaI - \Lambda^0_m NaBr = \Lambda^0_m I - \Lambda^0_m Br$

$\Lambda^0_m NaBr - \Lambda^0_m KBr = \Lambda^0_m Na - \Lambda^0_m K$

Q.12 The redox reaction among the following is :

(1) combination of dinitrogen with dioxygen at 200 K

(2) reaction of H_2SO_4 with NaOH

(3) formation of ozone from atmospheric oxygen in the presence of sunlight

(4) reaction of $[Co(H_2O)_6]Cl_3$ with $AgNO_3$

Ans. **[1]**

Sol. $N_2 + O_2 \longrightarrow 2NO$ (Ox. state charge)

$3O_2 \longrightarrow 2O_3$

$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

$AgNO_3 + NaCl \longrightarrow NaNO_3 + AgCl$

Q.13 For the reaction

$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$

the observed rate expression is, rate = $k_f[NO]^2[H_2]$. The rate expression for the reverse reaction is :

(1) $k_b[N_2][H_2O]^2/[H_2]$ (2) $k_b[N_2][H_2O]$ (3) $k_b[N_2][H_2O]^2/[NO]$ (4) $k_b[N_2][H_2O]^2$

Ans. **[1]**

Sol. $2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$

According to Rate expression forward

$2NO + H_2$ are Reactants

Hence one mole of H_2 is reacted in second step

$k_b[N_2][H_2O]^2/[H_2]$

Q.14 The bond order and the magnetic characteristics of CN^- are :

(1) $2\frac{1}{2}$, diamagnetic (2) 3, paramagnetic (3) $2\frac{1}{2}$, paramagnetic (4) 3, diamagnetic

Ans. **[4]**

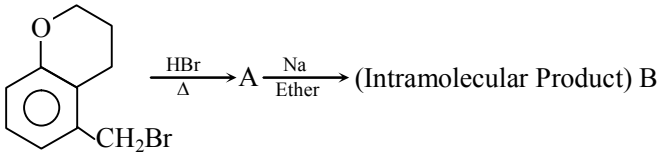
Sol. $CN^{\ominus}$ has 14 electron

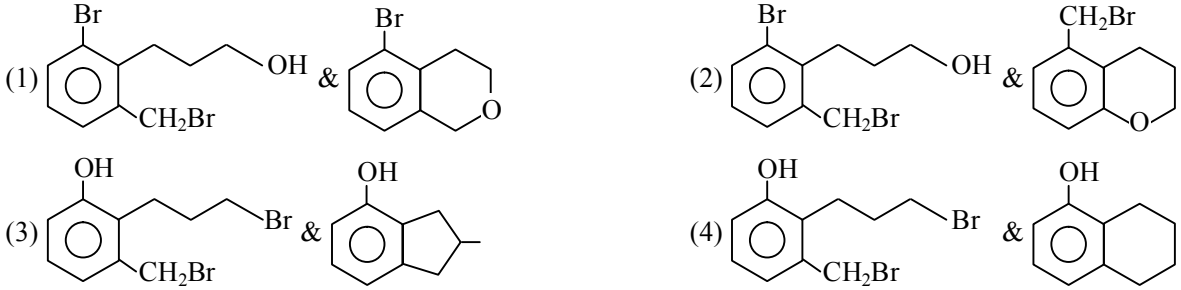
$\sigma 1s^2, \sigma^*1s^2\, \sigma 2s^2\, \sigma^*2s^2\, \pi^2 2py = \pi^2 2pz\, \pi^{*2}2py = \pi^{*2}2pz$

Bond order = $\frac{N_b - N_a}{2} = 3\ (C^{\ominus} \equiv N)$

diamagnetic $\rightarrow$ all electrons are paired

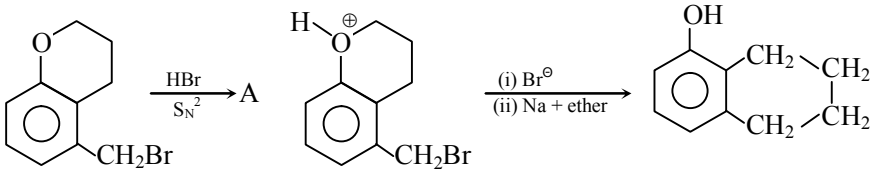
Q.15 In the following reaction sequence, structures of A and B, respectively will be :





Ans. **[4]**

Sol.



Q.16 Which of the following statements is correct ?

(1) Gluconic acid is obtained by oxidation of glucose with HNO_3

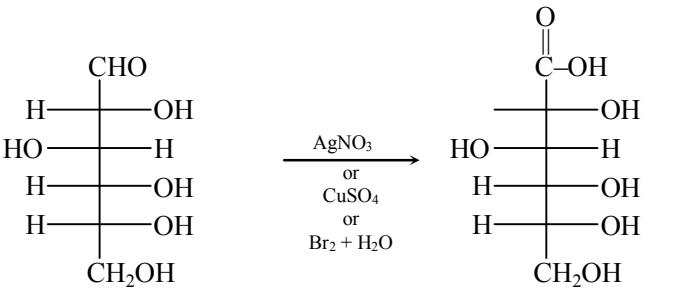
(2) Gluconic acid is a partial oxidation product of glucose

(3) Gluconic acid can form cyclic (acetal/hemiacetal) structure

(4) Gluconic acid is a dicarboxylic acid

Ans. **[2]**

Sol.



glucose gluconic acid

Q.17 Within each pair of element F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon and electron gain are :

(1) Cl, S and Li (2) F, Se and Na (3) Cl, Se and Na (4) F, S and Li

Ans. **[1]**

Sol.

Cl > F (exception)
S > Se (exception)
Li > Na (small size)

{3rd period > 2nd period}
For non metals

Q.18 The refining method used when the metal and the impurities have low and high melting temperatures, respectively, is :

(1) distillation (2) liquation
(3) zone refining (4) vapour phase refining

Ans. **[2]**

Sol. Theory Based

Q.19 The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is :

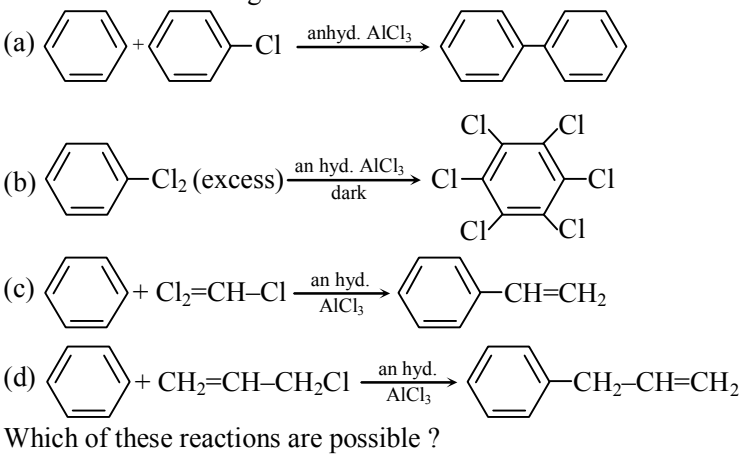
Note : A and B are unidentate neutral and unidentate monoanionic ligands, respectively.

(1) 0 and 1 (2) 0 and 0 (3) 0 and 2 (4) 2 and 2

Ans. **[2]**

Sol. MA_2B_2 shows geometrical isomerism not optical isomerism.

Q.20 Consider the following reaction :



Which of these reactions are possible ?

(1) (a) and (b) (2) (b) and (d) (3) (a) and (d) (4) (b), (c) and (d)

Ans. **[2]**

Sol. Vinyl halides and Aryl halides do not give Friedel Craft Reaction due to partial double bond character.

Q.21 The flocculation value of HCl for arsenic sulphide sol. is 30 m mol L^{-1}. If H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams, of H_2SO_4 in 250 ml required for the above purpose is _____.
(molecular mass of H_2SO_4 = 98 g/mol)

Ans. **[0.3675]**

Sol. $\rightarrow$ 1 litre solution of 30 m mol of HCl is required

$\rightarrow$ For 1 litre solution 15 m mol H_2SO_4 is required.

250 ml of solution required : - $\frac{15}{4} \times 10^{-3}$ (m mol) of H_2SO_4

= 0.3675 gm

Q.22 3 g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is _____.

[Given : pKa of acetic acid = 4.75, molar mass of acetic acid = 60 g/mol, log3 = 0.4771]
Neglect any changes in volume.

Ans. **[5.22]**

Sol. m mole of acetic acid in 20 ml = 2

m mole of HCl in 20 ml = 1

m mole of NaOH 20 ml = 2.5

$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$

(left)

1.5	2	0	0
0	0.5	1.5	

$pH = pKa + \log \frac{3/2}{2}$

$= 4.74 + \log 3 = 5.22$

Q.23 Consider the following reactions :

$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow (A) + \text{side products}$
(Conc.)

$(A) + NaOH \rightarrow (B) + \text{side products}$

$(B) + H_2SO_4 + H_2O_2 \rightarrow (C) + \text{Side products}$
(dilute)

The sum of the total number of atoms in one molecule each of (A), (B) and (C) is _____.

Ans. **[18]**

Sol. $NaCl \xrightarrow[H^{\oplus}]{K_2Cr_2O_7} \underset{(A)}{CrO_2Cl_2} \xrightarrow{NaOH} \underset{(B)}{NaCrO_4} \xrightarrow{H_2O_2/H^{\oplus}} CrO_5$

$CrO_2Cl_2 \rightarrow 5$

$NaCrO_4 \rightarrow 7$

$CrO_5 \rightarrow 6$

18 atoms

Q.24 The standard heat of formation $(\Delta_f H^0_{298})$ of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are –1560, –393.5 and –286 kJ/mol, respectively is _____.

Ans. **[192.5]**

Sol. $2C(\text{graphite}) + 3H_2(g) \longrightarrow 1C_2H_6 \qquad \Delta H_f(C_2H_6)$

(1) $C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O \qquad \Delta H_C = -1560$

(2) $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O \qquad \Delta H_C = -393.5$

(3) $C(\text{graphite}) + O_2 \longrightarrow CO_2 \qquad \Delta H_C = -286$ kJ/mol

$\Delta H_f(C_2H_6) = 3(-393.5) + 2(-286) + 1560$

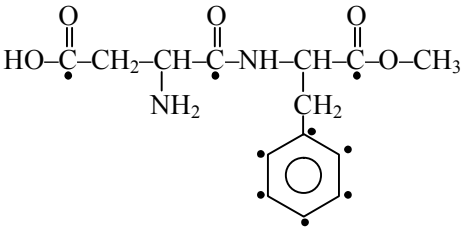
$= -1180.5 + (-572) + 1560$

$= 192.5$

Q.25 The number of sp^2 hybridised carbons present in "Aspartame" is ____.

Ans. **[9]**

Sol.



CAREER POINT

JEE Main Online Exam 2019

Questions & Answer

7th January 2019 | Shift - II

MATHS

Q.1 The number of ordered pairs (r, k) for which $6.^{35}C_r = (k^2 - 3).^{36}C_{r+1}$, where k is an integer, is:

(1) 6 (2) 3 (3) 2 (4) 4

Ans. **[4]**

Sol. $6.^{35}C_r = (k^2 - 3)\ ^{36}C_{r+1}$

$6.^{35}C_r = (k^2 - 3).\ \frac{36}{r+1}\ ^{35}C_r.$

$\Rightarrow k^2 - 3 = \frac{r+1}{6}$

$k \in I \Rightarrow r$ can be

(1) $r = 5 \Rightarrow k = \pm 2$

(2) $r = 35 \Rightarrow k = \pm 3$

4 ordered pairs (5, 2), (5, –2), (35, 3), (35, 3)

Q.2 The value of α for which $4\alpha\int_{-1}^{2} e^{-\alpha|x|}\ dx = 5$, is:

(1) $\log_e \sqrt{2}$ (2) $\log_e\left(\frac{3}{2}\right)$ (3) $\log_e\left(\frac{4}{3}\right)$ (4) $\log_e 2$

Ans. **[4]**

Sol. $4\alpha \int_{-1}^{2} e^{-\alpha|x|}\ dx = 5$

$4\alpha \int_{-1}^{0} e^{\alpha x} dx + \int_{0}^{2} e^{-\alpha x} dx\ = 5$

$\Rightarrow 4\alpha\left(\frac{e^{\alpha x}}{\alpha}\right)_{-1}^{0} + 4\alpha\left(\frac{e^{-\alpha x}}{-\alpha}\right)_{0}^{2} = 5$

$\Rightarrow 4\alpha\left(\frac{1-e^{-\alpha}}{\alpha} - \frac{e^{-2\alpha}-1}{-\alpha}\right) = 5$

Let $e^{-\alpha} = t$

$\therefore\ 4t^2 + 4t - 3 = 0$

$\Rightarrow t = \frac{1}{2}$

$e^{-\alpha} = \frac{1}{2}$

$\alpha = \ell n\ 2$

Q.3 Let f(x) be a polynomial of degree 5 such that x = ±1 are its critical points. If $\lim_{x\to 0}\left(2+\frac{f(x)}{x^3}\right) = 4$, then which one of the following is not ture?

(1) f is an odd function

(2) f(1) – 4f(–1) = 4

(3) x = 1 is a point of minima and x = –1 Is a point of maxima of f

(4) x = 1 is a point of maxima and x = –1 Is a point of minimum of f

Ans. **[4]**

Sol. f '(x) = 0 at x = 1, –1, also 0 is a repeated root.

$\therefore$ f '(x) = a (x + 1)(x – 1) x^2

= a(x^2 – 1)x^2

f'(x) = $ax^4 - x^2$

$f(x) = \frac{ax^5}{5} - \frac{ax^3}{3} + C$

$\because$ f(0) = 0 $\Rightarrow$ c = 0

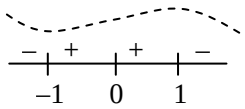
$\lim_{x\to 0}\frac{f(x)}{x^3} = 2$

$\Rightarrow 0 - \frac{a}{3} = 2$

$\Rightarrow$ a = –6

$\therefore f(x) = -\frac{6}{5}x^5 + 2x^3$

$\therefore$ f '(x) = –6(x^2 – 1)(x^2)



Minima at x = –1

Maxima at x = 1

Q.4 Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If $\lambda = \vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ and $\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$, then the ordered pair, $(\lambda,\vec{d})$ is equal to:

(1) $\left(-\frac{3}{2}, 3\vec{c}\times\vec{b}\right)$ (2) $\left(-\frac{3}{2}, 3\vec{a}\times\vec{b}\right)$ (3) $\left(\frac{3}{2}, 3\vec{b}\times\vec{c}\right)$ (4) $\left(\frac{3}{2}, 3\vec{a}\times\vec{c}\right)$

Ans. **[2]**

Sol. $\left|\vec{a}+\vec{b}+\vec{c}\right|^2 = 0$

$3 + 2\ (\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}) = 0$

$\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a} = -\frac{3}{2}$

$\lambda = -\frac{3}{2}$

$\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$

$= \vec{a}\times\vec{b}+\vec{b}\times(-\vec{a}-\vec{b})+(-\vec{a}-\vec{b})\times\vec{a}$

$\vec{d}=3(\vec{a}\times\vec{b})$

Q.5 Let y = y(x) be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$.

Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to:

(1) $\frac{2}{\sqrt{5}}$ (2) $\frac{\sqrt{5}}{2}$ (3) $-\frac{\sqrt{5}}{4}$ (4) $-\frac{\sqrt{5}}{2}$

Ans. **[4]**

Sol. $y.\sqrt{1-x^2} = k - x.\sqrt{1-y^2}$

On differentiating

$y.\frac{-2x}{2\sqrt{1-x^2}} + \sqrt{1-x^2}.y' = +\frac{x.2yy'}{2\sqrt{1-y^2}} - \sqrt{1-y^2}$

Put $x = \frac{1}{2}$, $y = -\frac{1}{4}$ & $x.y = -\frac{1}{8}$

On solving we get

$y' = \frac{-\sqrt{5}}{2}$

Q.6 The coefficient of x^7 in the expression $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + \ldots\ldots + x^{10}$ is:

(1) 210 (2) 120 (3) 330 (4) 420

Ans. **[3]**

Sol. $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + \ldots\ldots\ldots + x^{10}$

$$(1 + x)^{10}\frac{\left[1-\left(\frac{x}{1+x}\right)^{11}\right]}{\left(1-\frac{x}{1+x}\right)}$$

$\Rightarrow (1 + x)^{11} - x^{11}$

Coefficient of x^7 is $^{11}C_7 = 330$

Q.7 Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

(1) $p_5 = p_2 \cdot p_3$ (2) $p_3 = p_5 - p_4$

(3) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$ (4) $p_5 = 11$

Ans. **[1]**

Sol. $p_5 = \alpha^5 + \beta^5$

$\because \alpha^2 = \alpha + 1$

$= (\alpha + 1)^2.\alpha + (\beta + 1)^2.\beta$

$= (\alpha^2 + 2\alpha + 1)\alpha + (\beta^2 + 2\beta + 1)\beta$

$= (3\alpha + 2)\alpha + (3\beta + 2)\beta$

$= 3\alpha^2 + 2\alpha + 3\beta^2 + 2\beta$

$= 5\alpha + 5\beta + 6$ $[\because \alpha + \beta = 1]$

$= 5(1) + 6 = 11$

$p_2 = \alpha^2 + \beta^2 = \alpha + \beta + 2 = 3$

$p_3 = \alpha^3 + \beta^3 = (\alpha + 1).\alpha + (\beta + 1).\beta$

$= \alpha^2 + \beta^2 + \alpha + \beta$

$= \alpha + \beta + 3$

$= 1 + 3 = 4$

Hence $p_5 \neq p_2.p_3$

Q.8 If the sum of the first 40 terms of the series, 3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 +…… is (102)m, then m is equal to:

(1) 20 (2) 25 (3) 10 (4) 5

Ans. **[1]**

Sol. 3 + 4 + 8 + 9 + 13 + 14 +…….upto 40 terms

$\Rightarrow$ 7 + 17 + 27 +…….20 terms

$S = \frac{20}{2}[2 \times 7 + 19 \times 10]$

$= 102 \times 20 = 102$ m

$\therefore$ m = 20

Q.9 The area (in sq. units) of the region $\{(x, y) \in R^2 : 4x^2 \le y \le 8x + 12\}$ is:

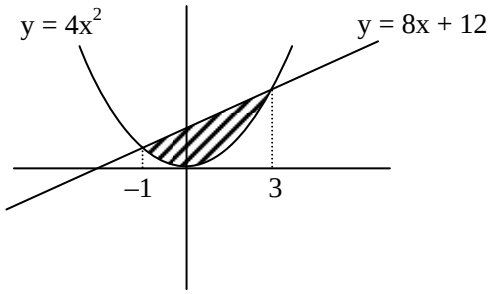
(1) $\frac{128}{3}$ (2) $\frac{127}{3}$ (3) $\frac{124}{3}$ (4) $\frac{125}{3}$

Ans. **[1]**

Sol. For point of intersection

$4x^2 = 8x + 12$

$x = -1, 3$



$\text{Area} = \int_{-1}^{3}[(8x+12) - 4x^2]\,dx$

$= \left[\frac{8x^2}{2} + 12x - \frac{4x^3}{3}\right]_{-1}^{3}$

$= (36 + 36 - 36) - (4 - 12 + \frac{4}{3})$

$= \frac{128}{3}$

Q.10 Let $a_1, a_2, a_3, \ldots$ be a G.P. such that $a_1 < 0$, $a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If $\sum_{i=1}^{9} a_i = 4\lambda$, then λ is equal to:

(1) $\frac{511}{3}$ (2) –171 (3) –513 (4) 171

Ans. **[2]**

Sol. $a_1 + a_2 = 4 \Rightarrow a_1 + a_1 r = 4$ ……(i)

$a_3 + a_4 = 16 \Rightarrow a_1 r^2 + a_1 r^3 = 16$ ……(ii)

equation (i) ÷ (ii)

$\Rightarrow \frac{1}{r^2} = \frac{1}{4}$

$\Rightarrow r = \pm 2$

$r = 2 \Rightarrow a_1 = \frac{4}{3}$

$r = -2 \Rightarrow a_1 = -4$

$\sum_{i=1}^{9} a_i = \frac{a(r^9 - 1)}{(r-1)} = (-4)\frac{((-2)^9 - 1)}{(-2-1)}$

$= \frac{4}{3}(-513) = 4\lambda$

$\Rightarrow \lambda = -171$

Q.11 If θ_1 and θ_2 be respectively the smallest and the largest values of θ in $(0, 2\pi) - \{\pi\}$ which satisfy the equation, $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$, then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta \, d\theta$ is equal to:

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{9}$ (3) $\frac{\pi}{3} + \frac{1}{6}$ (4) $\frac{2\pi}{3}$

Ans. **[1]**

Sol. $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$

$2\frac{\cos^2\theta}{\sin^2\theta} - \frac{5}{\sin\theta} + 4 = 0$

$2\sin^2\theta - 5\sin\theta + 2 = 0$

$(2\sin\theta - 1)(\sin\theta - 2) = 0$

$\sin\theta = \frac{1}{2}$ only

$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$\downarrow \quad \downarrow$

$\theta_1 \quad \theta_2$

$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \cos^2 3\theta \, d\theta = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \left(\frac{1+\cos 6\theta}{2}\right) d\theta$

$= \frac{1}{2}\left(\theta + \frac{\sin 6\theta}{6}\right)_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$

$= \frac{\pi}{3}$

Q.12 If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in R$, then the distance between the foci of the ellipse is:

(1) 4 (2) $2\sqrt{2}$ (3) $2\sqrt{5}$ (4) $2\sqrt{7}$

Ans. **[4]**

Sol. $3x + 4y = 12\sqrt{2}$

$y = -\frac{3x}{4} + 3\sqrt{2}$

line is tangent to ellipse

$\therefore\ c^2 = a^2m^2 + b^2$

$(3\sqrt{2})^2 = a^2\left(-\frac{3}{4}\right)^2 + 9$

$18 = \frac{9a^2}{16} + 9$

$2 = \frac{a^2}{16} + 1$

$a^2 = 16$

$\therefore\ e^2 = 1 - \frac{b^2}{a^2}$

$e^2 = 1 - \frac{9}{16}$

$e = \frac{\sqrt{7}}{4}$

Distance between focii = 2ae

$= 2 \times 4 \times \frac{\sqrt{7}}{4}$

$= 2\sqrt{7}$

Q.13 Let A = $[a_{ij}]$ and B = $[b_{ij}]$ be two 3 × 3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where i, j = 1, 2, 3. If the determinant of B is 81, then the determinant of A is:

(1) 3 (2) $\frac{1}{9}$ (3) $\frac{1}{81}$ (4) $\frac{1}{3}$

Ans. **[2]**

Sol. $|B| = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} = \begin{vmatrix} 3^0a_{11} & 3^1a_{21} & 3^2a_{31} \\ 3^1a_{12} & 3^2a_{22} & 3^3a_{32} \\ 3^2a_{13} & 3^3a_{23} & 3^4a_{33} \end{vmatrix}$

$81 = 3^3 \cdot 3^3 \cdot 3^2 |A|$

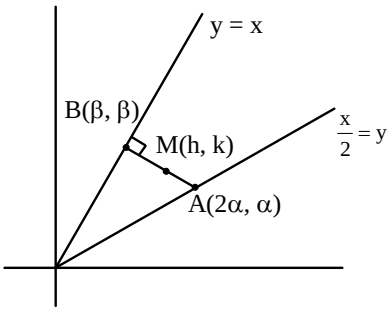
$\Rightarrow |A| = \frac{1}{9}$

Q.14 The locus of the mid-points of the perpendiculars drawn from points on the line, x = 2y to the line x = y is:

(1) 2x – 3y = 0 (2) 5x – 7y = 0 (3) 7x – 5y = 0 (4) 3x – 2y = 0

Ans. **[2]**

Sol.



Slope AB = $\frac{k-\alpha}{h-2\alpha} = -1$

$\Rightarrow \alpha = \frac{k+k}{3}$ …… (1)

also $\frac{\beta+2\alpha}{2} = h$, $\frac{\beta+\alpha}{2} = k$

$\alpha = 2h - 2k$ …… (2)

from (1) & (2)

$\frac{h+k}{3} = 2h - 2k$

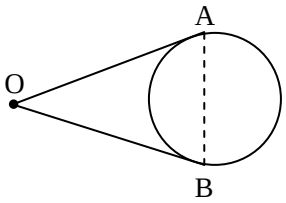
$\Rightarrow 5h = 7k$

$\Rightarrow 5x = 7y$

Q.15 Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to:

(1) $\frac{56}{5}$ (2) $\frac{64}{5}$ (3) $\frac{32}{5}$ (4) $\frac{52}{5}$

Ans. **[2]**



OA = $\sqrt{S_1}$

$\ell = \sqrt{16} = 4$

Radius = R = $\sqrt{16+4-16} = 2$

Length of AB = $\frac{2RL}{\sqrt{L^2+R^2}} = \frac{2\times4\times2}{\sqrt{16+4}} = \frac{16}{\sqrt{20}}$

$AB^2 = \frac{16\times16}{20} = \frac{64}{5}$

Q.16 The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is:

(1) $\frac{4-\sqrt{5}}{3}$ (2) $\frac{2}{3}$ (3) $\frac{\sqrt{7}-2}{3}$ (4) $\frac{4-\sqrt{7}}{3}$

Ans. **[4]**

$f(x) = x^3 - 4x^2 + 8x + 11$, $x \in [0, 1]$

Using LMVT

$f'(c) = \frac{f(1)-f(0)}{1-0}$

$3c^2 - 8c + 8 = \frac{16-11}{1-0}$

$3c^2 - 8c + 3 = 0$

$c = \frac{4-\sqrt{7}}{3} \in (0, 1)$

Q.17 Let A, B, C and D be four non-empty sets. The contrapositive statement of "If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$" is:

(1) If $A \not\subseteq C$, then $A \subseteq B$ and $B \subseteq D$ (2) If $A \not\subseteq C$, then $A \not\subseteq B$ and $B \subseteq D$

(3) If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$ (4) If $A \subseteq C$, then $B \subset A$ or $D \subset B$

Ans. **[3]**

Sol. If $A \subseteq B$ and $B \subseteq D$ then $A \subseteq C$

Contrapositive is

If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$

Q.18 If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta \in [0, 2\pi]$, is a real number, then an argument of $\sin\theta + i\cos\theta$ is:

(1) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$ (2) $\pi - \tan^{-1}\left(\frac{4}{3}\right)$ (3) $\tan^{-1}\left(\frac{4}{3}\right)$ (4) $-\tan^{-1}\left(\frac{3}{4}\right)$

Ans. **[2]**

Let $z = \frac{3+i\sin\theta}{4-i\cos\theta} \times \frac{(4+i\cos\theta)}{(4+i\cos\theta)}$

$= \frac{12 - \sin\theta\cos\theta + i(4\sin\theta + 3\cos\theta)}{16 + \cos^2\theta}$

z is real

$\therefore$ $4\sin\theta + 3\cos\theta = 0$

$\Rightarrow \tan\theta = \frac{-3}{4}$ [$\because$ θ lies is 2nd quadrant]

$\arg(\sin\theta + i\cos\theta) = \pi + \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right)$

$= \pi - \tan^{-1}\left(\frac{4}{3}\right)$

Q.19 In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3$ k, then k is equal to:

(1) 4 (2) $\frac{17}{4}$ (3) $\frac{17}{8}$ (4) $\frac{17}{2}$

Ans. **[3]**

Prob. (at most two machines will be out of service) = $\left(\frac{3}{4}\right)^3$.k

$\Rightarrow {}^5C_0\left(\frac{3}{4}\right)^5 + {}^5C_1\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^4 + {}^5C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3 = \left(\frac{3}{4}\right)^3.k$

$\Rightarrow \frac{17}{8}.\left(\frac{3}{4}\right)^3 = \left(\frac{3}{4}\right)^3.k$

$\Rightarrow k = \frac{17}{8}$

Q.20 Let y = y(x) be the solution curve of the differential equation, $(y^2 - x)\frac{dy}{dx} = 1$, satisfying y(0) = 1. This curve intersects the x-axis at a point whose abscissa is:

(1) 2 (2) 2 + e (3) 2 – e (4) –e

Ans. **[3]**

$\frac{dx}{dy} = y^2 - x$

$\frac{dy}{dx} + x = y^2$

I.F. = $e^{\int 1dy} = e^y$

$\Rightarrow x.e^y = \int y^2.e^y dy$

$x.e^y = y^2.e^y - \int 2y.e^y dy$

$xe^y = y^2e^y - 2ye^y + 2e^y + C$

$\because$ y(0) = 1

$\Rightarrow C = -e$

$\therefore xe^y = y^2e^y - 2ye^y + 2e^y - e$

put y = 0

$\therefore x = 0 - 0 + 2 - e$

$\Rightarrow x = 2 - e$

Q.21 Let X = {n ∈ N : 1 ≤ n ≤ 50}. If A = {n ∈ X : n is a multiple of 2} and B = {n ∈ X : n is a multiple of 7}, then the number of elements in the smallest subset of X containing both A and B is_____.

Ans. **[29]**

Sol. A = $\{2, 4, 6, 8.....50\} \Rightarrow$ 25 elements

A = $\{7, 14, 21, ...49\} \Rightarrow$ 7 elements

$A \cap B = \{14, 28, 42\}$ = 3 elements

Required number of elements = 25 + 7 – 3 = 29

Q.22 If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then x.y is equal to_____.

Ans. **[54]**

Sol. Means = $10 = \frac{3+7+9+12+13+20+x+y}{8}$

16 = x + y ___(1)

Variance $\sigma^2 = 25 = \frac{\sum x_i^2}{8} - (\text{mean})^2$

$25 = \frac{3^2+7^2+9^2+12^2+13^2+20^2+x^2+y^2}{8} - 100$

$125 \times 8 = 9 + 49 + 81 + 144 + 169 + 400 + x^2 + y^2$

$x^2 + y^2 = 148$ ___(2)

$(x + y)^2 = x^2 + y^2 + 2xy$

$256 = 148 + 2xy$

x.y = 54

Q.23 If the function f defined on $\left(-\frac{1}{3},\frac{1}{3}\right)$ by $f(x) = \begin{cases} \frac{1}{x}\log_e\left(\frac{1+3x}{1-2x}\right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$ is continuous, then k is equal to_____.

Ans. **[5]**

Sol. $\lim\limits_{x\to 0} f(x) = \lim\limits_{x\to 0}\left(\frac{1}{x}\ell n(1+3x) - \frac{1}{x}\ell n(1-2x)\right)$

$= \lim\limits_{x\to 0}\left(\frac{3\ell n(1+3x)}{3x} - \frac{2\ell n(1-2x)}{-2x}\right)$

$= 3 + 2 = 5$

f is continuous $\therefore \lim\limits_{x\to 0} f(x) = f(0)$

$\therefore f(0) = 5 = k$

Q.24 If the system of linear equations,

$x + y + z = 6$

$x + 2y + 3z = 10$

$3x + 2y + \lambda z = \mu$

has more than two solutions, then $\mu - \lambda^2$ is equal to_____.

Ans. **[]**

Sol. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 2 & \lambda \end{vmatrix} = 0$

$\Rightarrow 1(2\lambda - 6) - 1(\lambda - 9) + 1(-4) = 0$

$\Rightarrow 2\lambda - 6 - \lambda + 9 - 4 = 0$

$\Rightarrow \lambda = 1$

$\Delta_x = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 3 \\ \mu & 2 & \lambda \end{vmatrix} = 0$

$\Rightarrow 6(2\lambda - 6) - 1(10\lambda - 3\mu) + 1(20 - 2\mu) = 0$

$\Rightarrow 12\lambda - 36 - 10\lambda + 3\mu + 20 - 2\mu = 0$

$\Rightarrow 2\lambda + \mu = 16$

$\Rightarrow 2 + \mu = 16$

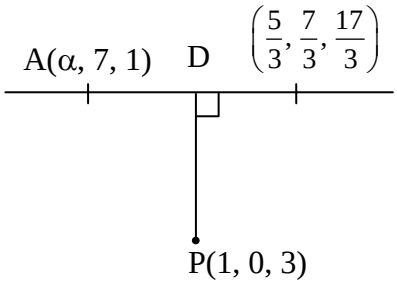
$\Rightarrow \mu = 14$

$\mu - \lambda^2 = 14 - 1 = 13$

Q.25 If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through (α, 7, 1) is $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$, then α is equal to_____.

Ans. **[4]**

Sol. $\Delta \equiv \left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$



$\overrightarrow{AD}.\overrightarrow{PD} = 0$

$$\left(\left(\frac{5}{3}-\alpha\right)\hat{i}+\left(\frac{7}{3}-7\right)\hat{j}+\left(\frac{17}{3}-1\right)\hat{k}\right).\left(\frac{2}{3}\hat{i}+\frac{7}{3}\hat{j}+\frac{8}{3}\hat{k}\right) = 0$$

$$\left(\frac{5}{3}-\alpha\right)\frac{2}{3}+\frac{7}{3}\times\left(\frac{-14}{3}\right)+\frac{14}{3}\times\frac{8}{3} = 0$$

$2(5 - 3\alpha) - 14 \times 7 + 14 \times 8 = 0$

$5 - 3\alpha - 49 + 56 = 0$

$3\alpha = 12$

$\alpha = 4$